

FACULTY OF INFORMATION TECHNOLOGY
BACHELOR OF BUSINESS INFORMATION TECHNOLOGY
BBT 3103 – NETWORK ROUTING PROTOCOLS

Takeaway Assignment 1-Configuring Static Routing

Date: 17th June 2026

Instructions

- This assignment is due on **Friday 26th June 11:59pm EAT**.
- You will work in teams of two.
- Your complete assignment will include a **1-page PDF document (or Scanned PDF of your handwritten answers)** and a **packet tracer file**.
- The 1-page document should bear your **admission numbers and names on every page**.

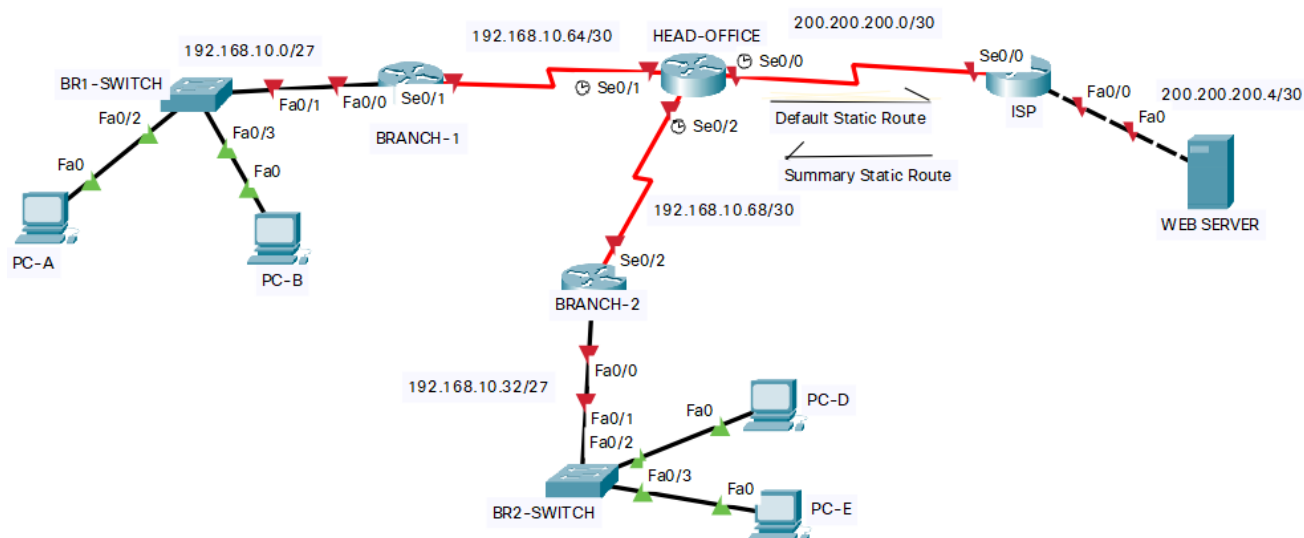
Assignment Objectives

At the end of the assignment you should be able to;

- Allocate appropriate IPv4 addresses to your devices
- Configure various static routes
- Troubleshoot various static configuration errors

The Scenario

A company connects to its branches using point-to-point WAN links as shown the topology below.



Part 1: Preparation

Examine the topology and determine which IP addresses you will use for the router ports and the PCs.

- Note that the subnetting has already been done. Below are the subnets as per the topology. Yours is to determine the addresses to allocate

Table 1

Router Connection	Subnet	Prefix
Branch 1-LAN	192.168.10.0	/27
Branch 2-LAN	192.168.10.32	/27
Branch 1 to Head-Office	192.168.10.64	/30
Branch 2 to Head-Office	192.168.10.68	/30
Head-Office-ISP	200.200.200.0	/30
Head-Office-LAN	200.200.200.4	/30

Refer to Table 1. Determine the IP addresses to be allocated and document your IPs in the table below

Interface/Device	IP Address	Decimal Mask	Default Gateway (where applicable)
Branch 1 – fa0/0			
PC-A			
PC-B			
Branch 2 – fa0/0			
PC-D			
PC-E			
Branch 1-S0/1			
Head-Office-S0/1			
Branch 2-S0/2			
Head-Office-S0/2			
Head-Office-S0/0			
ISP-S0/0			
ISP-Fa0/0			
Web-Server			

Part 2: Device Connections

Using the device guide below and the topology on page 1, connect your devices for this topology

Device	Model	Location
Routers	2621-XM	Miscellaneous Folder
Switches	2960	Network Devices
PCs	PC-PT	End Devices
Web Server	Server-PT	End Devices

Part 3: Basic Device Configurations

Using your IP Table in **Part 1** above, and the topology, perform the following configurations

- Router Names
- IP settings on routers *i.e. IP address, subnet mask, no shutdown and a clock rate of 128000 on the appropriate interfaces*
- IP Settings on the PCs and Webserver
- Ping to test for connectivity between the **directly connected devices**. *The ping between directly connected interfaces should succeed*
- Save the file as **Your Adm No-Name-Static e.g. 12345-Jane Doe-Static**
- Save the file as again as Your Adm No-Name-RIP e.g. 12345-Jane Doe-RIP

Note: The file in Part 3 (f) will be used in the next assignment to configure RIP

Part 4: Configure Static Routing on the file saved in (e) above

- a) Examine the routing table of Branch-1. List of routes that are directly connected. These are the routes that are currently known to the Branch1-Router.
- b) Examine the routing table of Branch-2. List of routes that are directly connected. These are the routes that are currently known to the Branch2-Router.
- c) Compare the output in (a) and (b) above. Take note of the routes that are on Branch-1 but are missing on Branch-2 and those on Branch-2 but are missing on Branch-1
- d) On Branch-1 configure static routes to reach the missing routes identified in (c) above
- e) On Branch-2 configure static routes to reach the missing routes identified in (c) above
- f) Examine the routing tables of both Branch-1 and Branch -2 again. The routing tables should now have the same networks.
- g) On Head-Office, configure static routes to reach the LANs on the branches.
- h) Examine the routing tables of Head-Office, Branch-1 and Branch-2. They should all have the same 192.168.10.x networks.
- i) From the PCs in Branch-1 ping the PCs on Branch-2. All pings should work. In case of a failure, troubleshoot and correct the errors.
- j) On Head-Office, configure a default static route pointing to the ISP; to reach the networks beyond the ISP
- k) On Branch-1, configure a default static route pointing to the Head-Office router to reach the networks not listed in Branch-1's routing table
- l) On Branch-2, configure a default static route pointing to the Head-Office router to reach the networks not listed in Branch-2's routing table
- m) Summarize all the 192.168.10.x networks on the topology. *Show your working*
- n) On ISP, configure the summary route in (m) above. Your static route command should ensure that the summary route points towards the Head-Office router.
- o) Gradually, test for connectivity between devices across the entire domain, troubleshoot and correct the errors *(if any)*

Note: *If well configured, all the devices should be able to ping with each other by the end of the exercise.*

Part 5: Submit your work

- a) Your complete assignment will include a **1-page PDF document** and a **packet tracer file**.
- b) Save your answers to **Part 1** and **Part 4 (m)** in a word document and convert to PDF or handwrite, scan and PDF your handwritten work. The document should bear your **admission numbers** and **names** on **every page**
- c) For the packet tracer file, indicate your **admission numbers** and **names** in a text box inside the packet tracer workspace.
- d) Ensure that the packet tracer file is saved using the format indicated in Part 3 (e) above.
- e) You will **upload your PDF and packet tracer file with static routing**, on or before the deadline.

Note: *Do not Zip the files. You may edit the submission if the assignment deadline has not expired.*